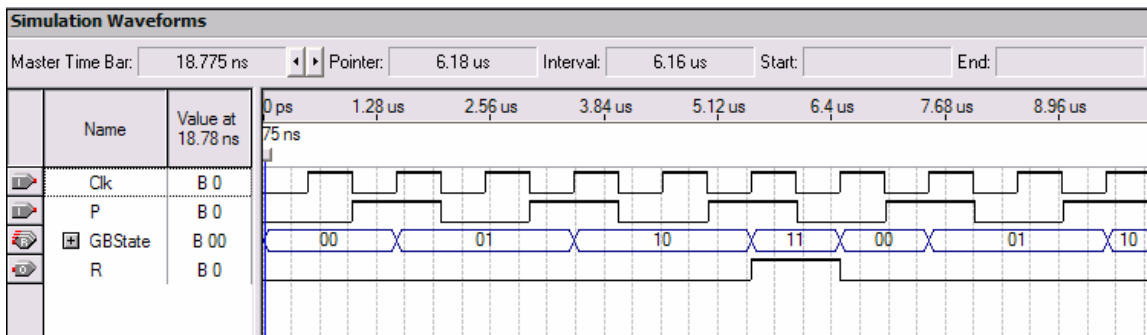


State Machines with VHDL

Simple Moore Architecture with User Defined States

```
ENTITY VHDLGumball IS
    PORT (      Clk, P      :      IN      STD_LOGIC;
           R          :      OUT      STD_LOGIC);
END VHDLGumball;

ARCHITECTURE SimpleMoore OF VHDLGumball IS
    SIGNAL GBState : STD_LOGIC_VECTOR (1 DownTo 0);
    CONSTANT A: STD_LOGIC_VECTOR (1 DownTo 0) := "00";
    CONSTANT B: STD_LOGIC_VECTOR (1 DownTo 0) := "01";
    CONSTANT C: STD_LOGIC_VECTOR (1 DownTo 0) := "10";
    CONSTANT D: STD_LOGIC_VECTOR (1 DownTo 0) := "11";
BEGIN
    PROCESS (Clk)
    BEGIN
        IF Clk'EVENT AND Clk='1' THEN
            CASE GBState IS
                WHEN A =>
                    IF P='1' THEN
                        GBState <= B;
                    ELSE
                        GBState <= A;  --Not necessary - implied memory.
                    END IF;
                WHEN B =>
                    IF P='1' THEN
                        GBState <= C;
                    END IF;
                WHEN C =>
                    IF P='1' THEN
                        GBState <= D;
                    END IF;
                WHEN D=>
                    IF P='1' THEN
                        GBState <= B;
                    ELSE
                        GBState <= A;
                    END IF;
            END CASE;
        END IF;
    END PROCESS;
    R <= '1' WHEN GBState=D ELSE '0';
END SimpleMoore;
```

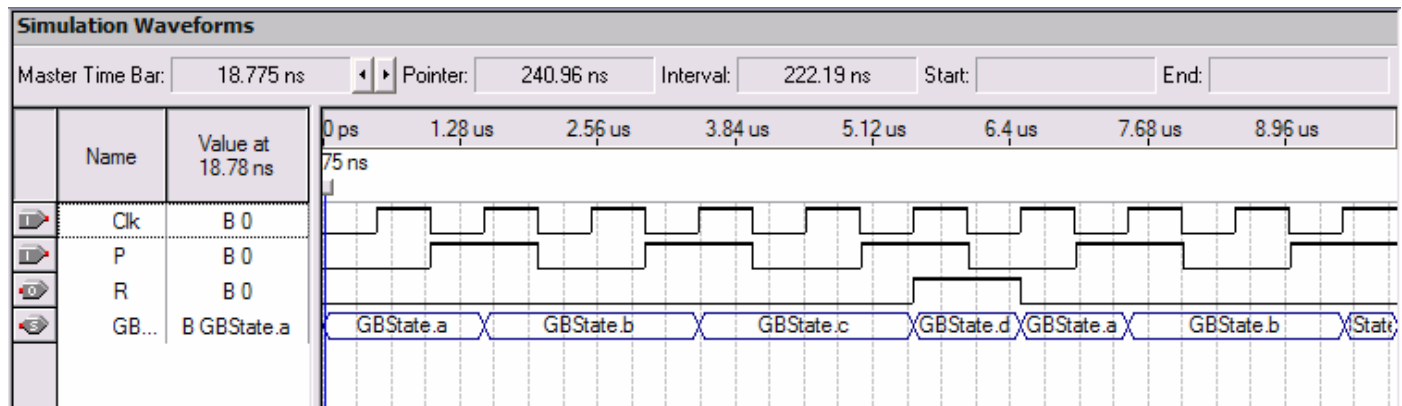


Moore Architecture with VHDL Defined States

```

ARCHITECTURE BetterMoore OF VHDLGumball IS
    TYPE GBState_type IS (A, B, C, D);
    SIGNAL GBState : GBState_Type;
BEGIN
    PROCESS (Clk)
    BEGIN
        IF Clk'EVENT AND Clk='1' THEN
            CASE GBState IS
                WHEN A =>
                    IF P='1' THEN
                        GBState <= B;
                    END IF;
                WHEN B =>
                    IF P='1' THEN
                        GBState <= C;
                    END IF;
                WHEN C =>
                    IF P='1' THEN
                        GBState <= D;
                    END IF;
                WHEN D=>
                    IF P='1' THEN
                        GBState <= B;
                    ELSE
                        GBState <= A;
                    END IF;
            END CASE;
        END IF;
    END PROCESS;
    R <= '1' WHEN GBState=D ELSE '0';
END BetterMoore;

```

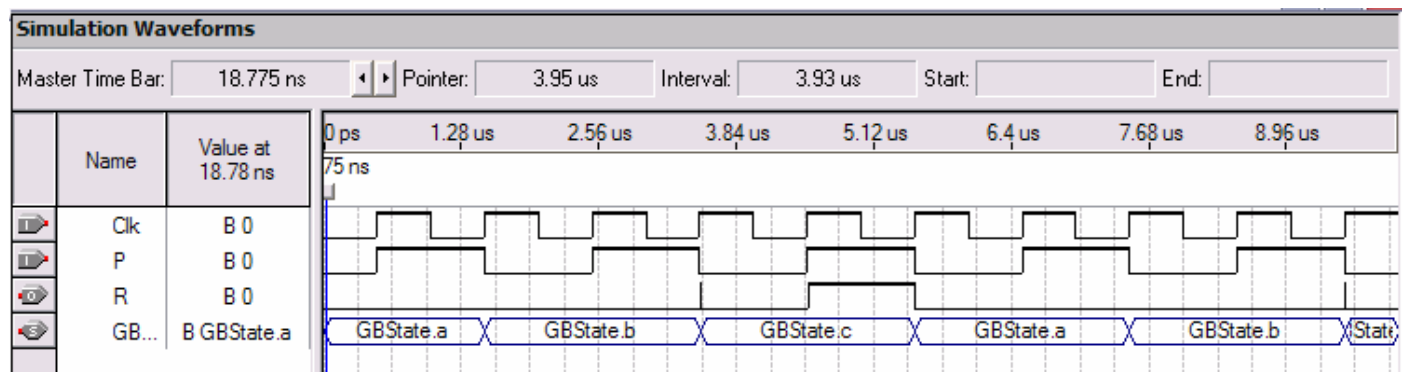
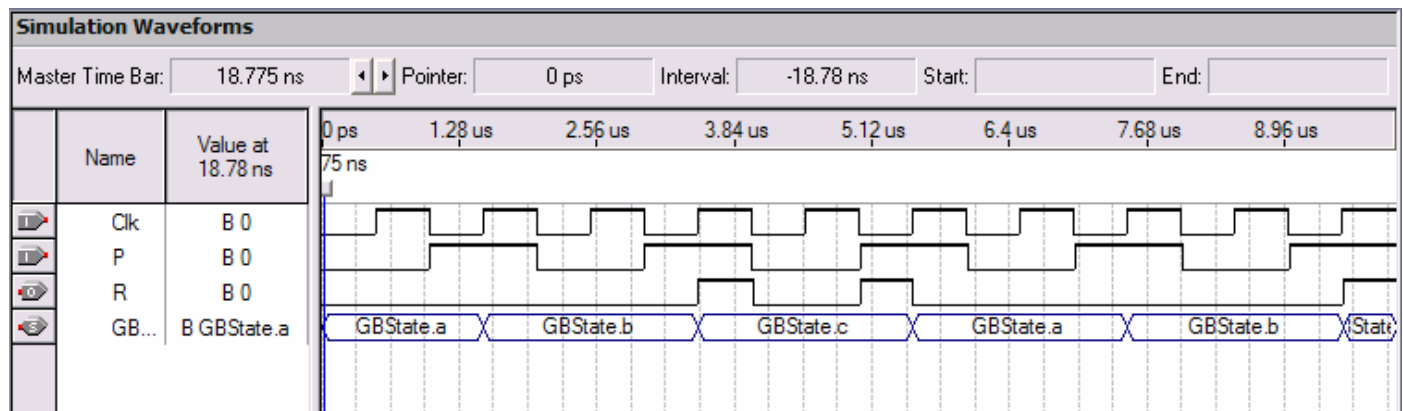


Mealy Architecture with VHDL Defined States

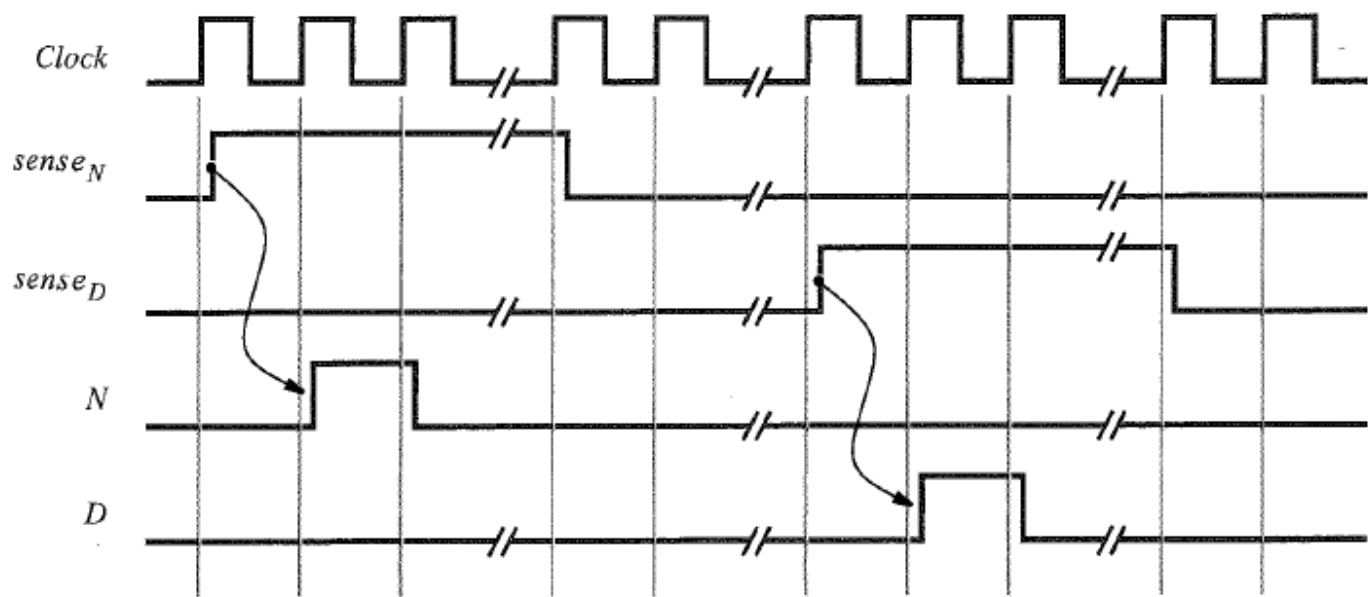
```

ARCHITECTURE Mealy OF VHDLGumball IS
    TYPE GBState_type IS (A, B, C);
    SIGNAL GBState : GBState_Type;
BEGIN
    PROCESS (Clk, P)
    BEGIN
        IF Clk'EVENT AND Clk='1' THEN
            CASE GBState IS
                WHEN A =>
                    IF P='1' THEN
                        GBState <= B;
                    END IF;
                WHEN B =>
                    IF P='1' THEN
                        GBState <= C;
                    END IF;
                WHEN C =>
                    IF P='1' THEN
                        GBState <= A;
                    END IF;
            END CASE;
        END IF;
    END PROCESS;
    R <= '1' WHEN (GBState=C AND P='1') ELSE '0';
END Mealy;

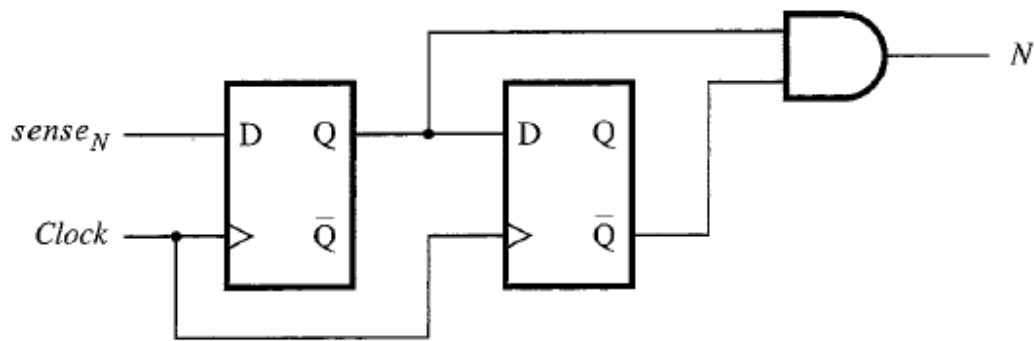
```



Sensing a coin (From textbook)



(a) Timing diagram



(b) Circuit that generates N

Figure 8.53 Signals for the vending machine.